Cover Page

SMART AI TRASH CAN

Auto Interceptor • Compactor • Sealing Bin

A conversational AI-powered waste management robot

THEME	Smart Products / AI / IoT • Sustainability & Waste Management
TEAM NAME	**HACTICKS**
TEAM ID	To be confirmed on the submission portal
STATE / UT	Tamil Nadu
PROJECT STATUS	Concept / proposed prototype — features are planned, not yet validated

HACTICKS

Problem – Proposed Solution

THE PROBLEM

Mixed waste, unhygienic contact and overflowing bins make everyday disposal harder.

1 HYGIENE

Frequent lid contact
Manual bag handling
Odour and overflow concerns

2 SEGREGATION

Biodegradable and non-biodegradable waste can get mixed
Sorting depends on user behaviour

3 CONVENIENCE

No interaction or fill feedback
Multiple bins take space
Disposal can be inconvenient

PROPOSED SOLUTION | A voice-enabled mobile bin with planned AI sorting, touch-free lid control, compaction and bag-sealing alerts.

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Technical Approach

Hack for Social Cause

PROPOSED SYSTEM ARCHITECTURE

SENSE	PROCESS	DECIDE	ACT	ALERT
Camera • ultrasonic IR / gas sensor	Raspberry Pi 4 + ESP32 controller	Voice intent • object / waste recognition	Lid • wheels • press plate • sealing kit	Speaker • LED phone alert

INTERACTION MODES

Service mode: user raises a hand; bin approaches and opens

Catch mode: camera tracks a thrown object (advanced goal)

IMPLEMENTATION PLAN

Prototype with Arduino IDE / ESP32 and Raspberry Pi software

Build and test one function at a time; add AI and motion after safety tests

AI usage: AI-assisted concept/design/presentation. GitHub & demo: not available yet. All architecture items are proposed.

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Impact Potential

DESIGNED FOR CLEANER, EASIER WASTE DISPOSAL

PEOPLE

Touch-free lid operation
Voice assistance
Clear fill-status alerts

ENVIRONMENT

Supports source segregation
May reduce overflow and mess
Encourages better disposal habits

OPERATORS

Potential use in homes, offices, public spaces and institutions
Less frequent manual checking

TARGET STAKEHOLDERS

Households • campuses • offices • facility managers • municipal sanitation teams

Expected benefits are design goals; impact will need to be measured in prototype trials.

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Feasibility and Risk Analysis

Hack for Social Cause

FEASIBILITY & RISK PLAN

INDICATIVE BUDGET

₹10,500–₹21,500

Estimate from concept notes; verify prices

KEY BUILD CHALLENGES

Reliable camera tracking
Battery life and weight
Safe compaction force
Separate control of heater and motors

Motion & collision	Low-speed movement, obstacle sensing, emergency stop
Heater / bag sealing	Insulated enclosure, thermal cutoff, adult supervision and testing
Wet waste / electronics	Sealed compartments, cable protection and safe drainage
AI misclassification	Show confidence / allow manual override; test diverse waste samples

Feasibility status: proposed only; no prototype performance or 3× capacity gain has been validated.

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Team

TEAM COMPOSITION & PROJECT JOURNEY

TEAM ROLES — PROPOSED ASSIGNMENT

01	Project lead & integration	Parsini Shakthi VG (IoT)
02	Embedded systems & sensors	Tanushri V (IoT)
03	AI / computer vision & voice	Sakthe Priya V (ECE)
04	Mechanical design & safety	All members
05	Research, testing & presentation	All members

Team ID: add the official ID from the submission portal before submission.

CHALLENGES TO SOLVE

Integrating camera, motors and voice

Protecting users from moving parts and heat

Testing segregation accuracy and battery runtime

NEXT MILESTONES

Build a stationary touch-free lid prototype

Add fill sensing and alerts

Test sorting, mobility and sealing as separate modules

References

REFERENCE & RESEARCH STARTING POINTS

ESP32 / ESP32-CAM	Espressif official documentation	https://docs.espressif.com/
Raspberry Pi	Official documentation	https://www.raspberrypi.com/documentation/
Arduino IDE	Official software and getting started	https://docs.arduino.cc/
Ultrasonic sensing	HC-SR04 datasheet / supplier datasheet	Add the exact component datasheet used
AI / computer vision	Model documentation selected during implementation	Add the exact model and source
Waste management context	Use official CPCB / MoHUA / municipal data	Add verified statistic and citation before submission

Links are starting points, not claims of completed implementation. Verify and cite exact sources used.